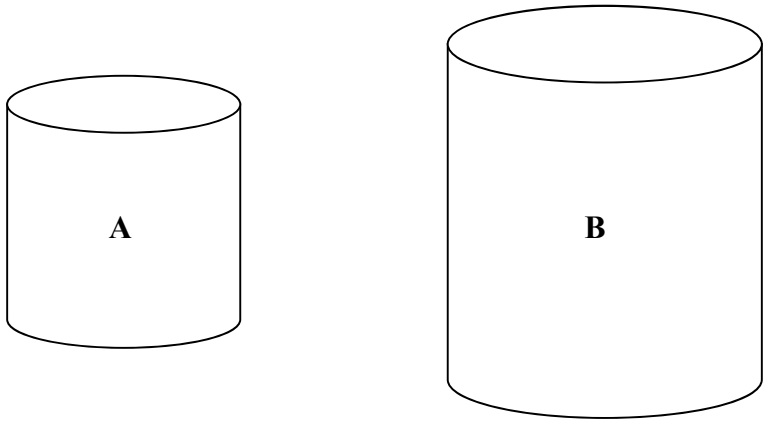
17 **A** and **B** are two similar cylindrical containers.



the surface area of container **A** : the surface area of container **B** = 4 : 9

Tyler fills container **A** with water.
She then pours all the water into container **B**.
Tyler repeats this and stops when container **B** is full of water.

Work out the number of times that Tyler fills container **A** with water.
You must show all your working.

Area ratio A : B = 4 : 9.

Length ratio A : B = √4 : √9 = 2 : 3.

Volume ratio A : B = 2^3 : 3^3 = 8 : 27.

B needs 27/8 = 3⅜ fills of A.

Three fills are insufficient; the 4th completes it.

4 marks total
Lengths 2 : 3
Volumes 8 : 27
27/8 = 3⅜
Round up to 4 fills

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(Total for Question 17 is 4 marks)